Multilayer REBCO Tape for 30-35 T Tokamak Magnets

MP35N Substrates, VOx Quench Layers, and Graphene-Aluminum Stabilizers for High-Field, Self-Protecting Coils

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Abstract

We present the design and performance of a new REBCO (Rare-Earth barium copper oxide) coated conductor tape engineered for ultra-high magnetic fields (30–35 T) in fusion tokamak magnets. The Kronos "S.M.A.R.T." REBCO tape employs a multilayer architecture with a high-strength metal substrate, textured oxide buffer layers, a thin superconducting REBCO film, and novel stabilizer enhancements. Key innovations include a MP35N alloy substrate to withstand large Lorentz forces, an integrated vanadium-oxide (VOx) interlayer for in situ quench protection, and graphene-reinforced aluminum elements embedded alongside copper stabilizer for improved high-field conductivity. The tape is designed around a strict strain limit (~0.4% tensile) to preserve superconducting performance under load[1][2]. We describe the tape's layer structure, fabrication processes (IBAD texturing, MOCVD growth, sputtering, electroplating), and mechanical support strategy. In magnet applications, the tape enables stand-alone REBCO coils to reach fields of 35 T – far beyond the ~23 T on-conductor record of present fusion magnets – while maintaining stability[3][4]. Performance metrics (critical current I_c, current density J_c, upper critical field Hc2, and thermal conductivity) are analyzed, demonstrating that the tape's enhancements push REBCO toward its intrinsic high-field limits. These advances support more compact, high-field tokamaks with robust, self-protecting magnets.

Table of Contents

Introduction 3
Tape Architecture 4
Fabrication Techniques 8
Mechanical Design and Lorentz Force Management 10
Quench Protection Strategy 14
Performance Metrics and Benchmarking 16
Risk Acknowledgment and Ongoing Research Disclosure 18
Conclusion 20
BIBLIOGRAPHY 22
TABLES 24
FIGURES 25

Introduction

High magnetic fields are a priority in fusion reactor design because stronger fields improve plasma confinement and allow smaller, more economical devices. Recent tokamak projects have adopted high-temperature superconductors (HTS) like REBCO to break the ~16 T field limit of traditional Nb_3Sn magnets. For example, MIT's SPARC tokamak plans to operate REBCO toroidal field (TF) coils at about 23 T peak field on conductor – a world record for a magnetically confined fusion device[3][4]. Kronos aims to push this further, targeting **30–35 T** in its compact S.M.A.R.T. (Superconducting, Minimum-Aspect-Ratio Tokamak) design[5][6]. Such extreme fields approach the intrinsic capability of REBCO: these ceramic superconductors have an upper critical field *Hc2* exceeding 70–100 T at cryogenic temperatures[1][7], vastly superior to low-temperature superconductors (e.g. Nb_3Sn's ~25 T limit at 1.8 K)[4][6]. In principle, REBCO can carry supercurrents at >30 T if kept cold (4–20 K) and properly reinforced[7][8].

The challenge is managing mechanical stress and strain in such high-field coils. REBCO tapes are intrinsically brittle; beyond ~0.4% tensile strain, the REBCO layer starts to irreversibly crack, causing critical current to drop precipitously[9][10]. Traditional REBCO tapes use a Hastelloy or Ni–W alloy substrate (~50–100 µm thick) that provides strength but can still be the weak link at very high stress[11][12]. Fusion magnets like SPARC mitigate this by massive external structures (steel clamps, epoxy impregnation) to limit tape strain[13][14]. Kronos instead adopts a "strain-first" design philosophy: begin with materials and architecture that inherently keep the REBCO under its strain limit, even at 35 T[15][16]. This motivated several key choices—starting with a higher-modulus substrate and adding internal reinforcement and quench safety layers—to create a tape that can operate near REBCO's performance envelope without damage[12][17].

REBCO was selected over other superconductors due to its superior high-field performance. Alternative HTS like MgB_2 or Fe-based superconductors have much lower upper critical fields (e.g. MgB_2 ~20 T at 4 K) and cannot reach 30+ T regimes[4]. Even the best low-temperature superconductor, Nb_3Sn, is limited to ~25 T and suffers degradation above ~0.2% strain[18][19]. REBCO's combination of high critical temperature (*T_c ~90 K*), high irreversibility field, and strong vortex pinning makes it the only viable conductor for 20–35 T magnet applications[20][21]. The Kronos tape leverages these intrinsic strengths of REBCO while addressing its weaknesses (mechanical brittleness and quench management) through an innovative multilayer structure. In the following, we detail the tape architecture (summarized in **Figure 1**), fabrication techniques, mechanical support strategy, quench protection approach, and the resulting performance characteristics. The goal is a robust **35 T-class HTS tape** that enables smaller, safer fusion magnets and unlocks new design possibilities beyond conventional superconductors.

Tape Architecture

Figure 1 illustrates the cross-sectional schematic of the Kronos S.M.A.R.T. REBCO tape, highlighting its multiple functional layers[22][23]. The tape is a few millimeters wide (typically 4–12 mm) and ~100 µm thick, built as a "coated conductor" with seven distinct layers engineered for specific roles. Below we list and describe each layer from the base substrate upward[24][25]:

1. **MP35N Support Substrate (≈50 µm):** A high-strength Ni-Co alloy strip forms the backbone of the tape, replacing the standard Hastelloy-C276 substrate. MP35N offers an exceptionally high tensile modulus (~230–234 GPa) and can sustain yield stresses up to ~1.5 GPa after work-hardening[26][27]. This stiff, tough base carries the mechanical load so that the brittle REBCO layer experiences minimal strain. Even under the enormous Lorentz forces of a 35 T magnet, the MP35N substrate limits the REBCO's tensile strain to <0.4%, preventing irreversible superconductor damage[25][28]. MP35N's cryogenic robustness and fatigue resistance also ensure the tape can be wound and repeatedly pulsed without cracking[29][30]. By "building in" mechanical strength, Kronos's tape can handle higher stress before needing external support[31][32].

Figure 1: Cross-section of MP35N substrate with superconducting ReBCO layer, showing the structural foundation used in HTS tapes for toroidal and poloidal coils.

2. **Textured Oxide Buffer Stack (~0.2 µm):** Above the metal substrate is a thin stack of oxide layers deposited by ion-beam assisted deposition (IBAD) and sputtering[33][34]. This typically includes layers like IBAD-MgO followed by YSZ or CeO_2. The buffer serves two critical purposes: (a) it acts as a chemical barrier to prevent metal diffusion from the substrate into the REBCO during high-temperature processing, and (b) it provides a biaxially textured template that guides the REBCO crystals to grow in alignment[35][34]. The result is a continuous, strongly textured REBCO film with high-quality grain boundaries. In essence, the buffer stack converts the random polycrystalline metal surface into an oriented, ceramic surface on which the superconductor can grow epitaxially[36][34]. This is crucial for achieving high critical current: the better the in-plane grain alignment, the higher the J_c the REBCO layer can carry across grain boundaries.

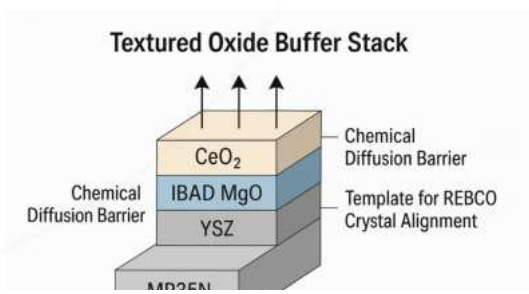

Figure 2: Textured oxide buffer stack on MP35N substrate, showing IBAD MgO, YSZ, and CeO_2 layers that block diffusion and template REBCO crystal growth.

3. **REBCO Superconducting Layer (~1.5–2 µm):** This is the heart of the tape – a thin film of rare-earth barium copper oxide ($REBa_2Cu_3O_{7-x}$, with RE = Y, Gd, etc.) that carries the supercurrent. The REBCO layer is deposited epitaxially on the buffered substrate, typically by a metal-organic chemical vapor deposition process (see Fabrication section). Despite its microscopic thickness, this layer carries extremely high current densities (on the order of 10^7–10^8 A/cm²) thanks to engineered flux pinning centers and excellent crystallinity[37][38]. Modern REBCO films (e.g. (Gd,Y)BCO with Zr doping) have demonstrated J_c > 20 MA/cm² at 30 K, 3 T in research settings[38][39]. In Kronos's tape, the REBCO is optimized to remain superconducting up to at least 35 T by operating at cryogenic temperatures (4–20 K) and maintaining low strain. Notably, REBCO's intrinsic irreversibility field can exceed 100 T at 4 K[40][1], so the 35 T operating point is still well below the material's theoretical field limit – the real constraints are mechanical, which the tape's design directly addresses.

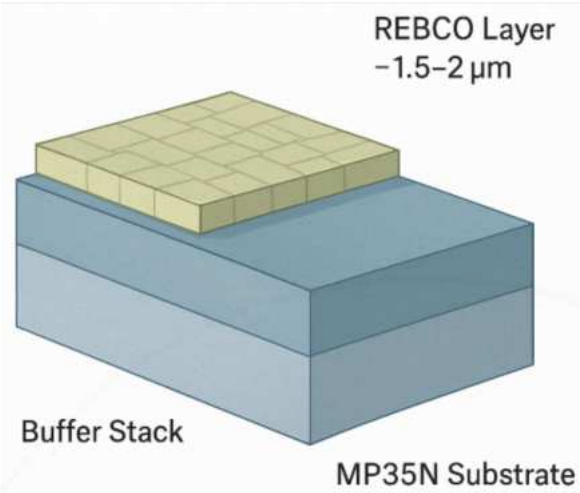

Figure 3: REBCO layer (~1.5–2 µm) on buffer stack with aligned grains enabling high current density for 30–35 T magnet operation.

4. **Silver Capping Layer (~1–2 µm Ag):** A thin silver layer is sputtered onto the REBCO film as a protective and functional cap. Silver is chemically inert and shields the superconductor from moisture or damage. It also provides an excellent low-resistance contact surface for electrical connections (current leads can be soldered to the Ag). Another important role is mechanical: silver is ductile, so this layer acts as a soft buffer that can redistribute localized stress and bridge any micro-cracks. If a small region of the REBCO tape transitions to the normal (resistive) state, the silver carries bypass current across that microscopic region until the copper stabilizer (added later) can take over. In short, the Ag overlayer improves the tape's robustness and connectivity without adding much thickness.

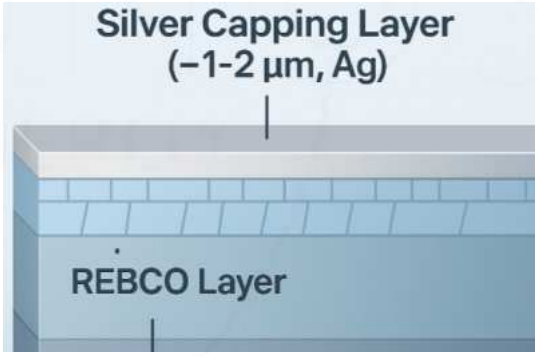

Figure 4: Cross-section showing the silver capping layer (~1–2 µm, Ag) atop the REBCO layer. This layer protects the

5. **Vanadium-Oxide Quench Layer (~1–2 µm V_2O_3):** A thin coating of vanadium oxide (specifically V_2O_3, a semiconducting oxide) is applied over the silver as a "smart" insulation layer[41][42]. This is a novel feature of the Kronos tape aimed at quench protection. At the cryogenic operating temperature (say 4–20 K), V_2O_3 is relatively resistive (especially when doped or oxygen-deficient to raise its transition temperature). Thus, during normal operation the layer acts as an insulator between turns, preventing significant inter-turn currents. However, if a local hot spot develops (quench), the temperature at that spot rises and the V_2O_3 undergoes a steep drop in resistivity[43][44]. In effect, the VOx layer becomes more metallic when heated. This property allows current to divert out of a quenching turn into adjacent turns once the local temperature rises, spreading the current and energy. By introducing a controlled turn-to-turn resistance that is temperature-sensitive, the VOx layer creates a self-protecting coil behavior[42][45]. It confines each turn's current during steady operation (minimizing eddy currents and ensuring each turn carries its share), but during a quench it permits prompt current sharing to avoid damage. This concept builds on the "no-insulation" (NI) coil technique developed for REBCO magnets[44], but with more control: the VOx provides a middle ground between fully insulated and fully uninsulated coils[46][47]. Kronos is an early adopter of this approach, which recent studies have shown can dramatically improve HTS magnet stability[47][48].

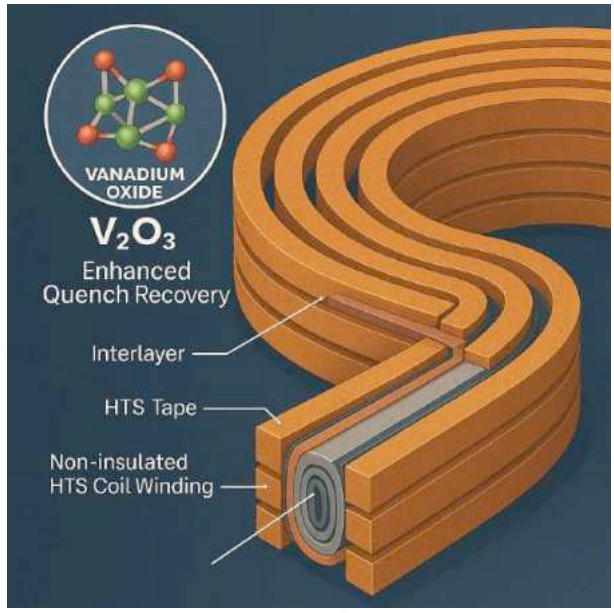

Figure 5: Layered HTS coil winding with non-insulated conductors, showing V_2O_3 interlayers for enhanced quench recovery and graphene-reinforced aluminum bypass paths.

6. **Graphene-Reinforced Aluminum Bypass (ultra-pure Al layers):** Uniquely, the Kronos tape embeds high-purity aluminum conductors, enhanced with graphene nanoreinforcement, alongside the superconducting core layers[49][50]. These appear as light-blue regions in **Figure 1** and effectively serve as parallel current paths (bypass strips) in case the REBCO goes normal. Aluminum is an excellent conductor with very high thermal conductivity at cryogenic temperatures. By using a very pure grade (with residual resistivity ratio RRR > 1000) and infusing graphene, Kronos achieves an aluminum layer that maintains low resistivity even in strong magnetic fields ~30+ T[51][50]. In contrast, copper's resistivity rises significantly at high fields due to magnetoresistance, which can hinder current sharing in >25 T magnets[52][50]. The

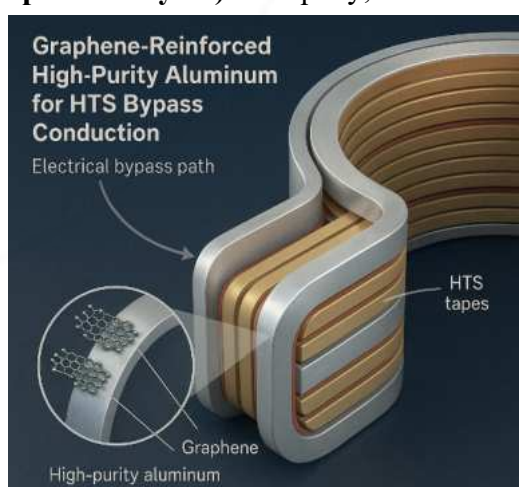

Figure 6: Section view of graphene-reinforced high-purity aluminum used for HTS bypass conduction, shown with stacked HTS tapes and carbon-fiber support plates.

graphene-enhanced Al bypass retains ~2×10^4 S/cm conductivity at cryogenic temperatures in 30–35 T fields[50][53], far better than copper in that regime. These aluminum strips are placed for optimal thermal and electrical contact with the REBCO layer. In a quench or fast ramp, they can quickly take up current and spread heat, reducing hot spots. Additionally, aluminum's thermal contraction is closer to REBCO's, which helps mitigate thermal stress on cooldown (the graphene reinforcement further tunes the CTE)[54][55]. Incorporating Al does complicate manufacturing (aluminum is not trivial to plate onto tape), but Kronos likely bonds or co-deposits these strips and then encapsulates them in copper (next layer)[56][57]. The payoff is a composite stabilizer where copper and aluminum together ensure low resistance from 0 T up to 35 T[58][59]. The Al handles the extreme field regime, while Cu handles the rest[60][58].

7. **Copper Stabilizer (~50 µm electroplated Cu):** Lastly, the entire tape is coated in thick high-purity copper, typically by electroplating[61][62]. Copper serves as the primary stabilizer, carrying the current in the event the superconductor quenches and providing a large thermal mass to absorb energy. In the Kronos tape, about 50 µm of copper is added, often on the top and bottom surfaces, effectively enveloping the tape in Cu[63][64]. The underlying silver layer ensures good electrical contact between REBCO and this copper. During a quench, the Cu stabilizer carries the diverted current and spreads heat over a broad area, preventing localized burnout[63][65]. Copper's RRR at cryogenic temperatures is extremely high (low resistivity), so it provides an effective shunt for the supercurrent. In Kronos's design, the copper works in tandem with the aluminum: at low to moderate fields the copper carries most of any normal current, but at very high fields where Cu's resistivity rises, the Al can take on more of the current[60][58]. Together they guarantee a low-resistance path at all field levels, enhancing stability up to 35 T. The copper coating also makes the tape more rugged and easier to handle, and allows the tape to be soldered into coil assemblies[63][66].

Figure 7: Cross-section highlighting the copper stabilizer (~50 µm) above the VO_x quench layer. This thick layer provides thermal mass and a low-resistance current-sharing path during a quench.

Advanced Materials				
Kronos Design Feature	**Value Proposition / Innovation**	**Quantitative Performance**	**Reference Institution**	**Peer-Reviewed Source**
Low-Activation Alloys	Reduces radioactive waste classification and disposal costs	D–^{3}He structures qualify as Class A waste vs D–T Class C	UW–Madison FTI	Safety & Environmental Characteristics of D–^{3}He Reactors (1992)
REBCO High-Tc Superconductors	Enables stronger fields and smaller reactors	20 T large-bore magnet demo (MIT, 2021)	MIT PSFC / CFS	MIT News: 20-T HTS Magnet Breakthrough
Carbon Fiber Reinforcement	Provides high stiffness with low activation	Toray M46J modulus ≈ 430 GPa	Toray Industries	Toray M46X Launch Release
Plasma-Facing Tungsten & Ceramics	Withstands extreme plasma heat flux	Tungsten divertors handle 10–20 MW/m²	ITER	ITER Divertor Materials

Table 1: Kronos geometry leverages spherical tokamak benefits (high β, compact design), along with negative triangularity for stability. The coil system stays under strain limits for REBCO integrity, supported by ultra-rigid carbon fiber structures.

In summary, the Kronos REBCO tape integrates multiple layers (substrate, buffer, REBCO, Ag, VOx, Al, Cu) to address the multifaceted demands of a fusion magnet conductor. Each layer is engineered for a role: the substrate and reinforcements bear mechanical loads, the REBCO carries supercurrent, and the cap and stabilizer layers protect and stabilize the tape under both normal and fault conditions. This multilayer approach yields a tape that is intrinsically more robust against strain and quenches than standard REBCO tapes[67][48].

Fabrication Techniques

Producing the above multilayer tape requires a sequence of advanced thin-film deposition and metallurgical processes[68][69]. Kronos's fabrication combines state-of-the-art methods from the commercial 2G HTS (second-generation high-temperature superconductor) industry – such as IBAD texturing and MOCVD growth – with custom steps for the novel materials (MP35N, VOx, graphene–Al). The major steps are outlined below:

Figure 8: Simplified process flow for REBCO tape fabrication, showing key steps: buffer deposition, REBCO growth, silver capping, VO_x layer application, and copper stabilizer electroplating.

- **Substrate Preparation:** The process begins with the MP35N substrate strip (~50 µm thick). MP35N foil can be manufactured by conventional cold working and annealing, or possibly electroforming for ultra purity[70][71]. The substrate is polished to a mirror finish, since a smooth surface is needed for subsequent buffer deposition. In some cases, an initial thin layer of metal (e.g. Cu or Ni) may be electrodeposited to further smooth the substrate[71]. Kronos's internal diagrams hint that the substrate could be electroformed to achieve a clean, defect-free foil[71], but the key distinction is using MP35N in place of Hastelloy.
- **Buffer Layer Deposition (IBAD & sputtering):** Next, the textured oxide buffer stack is grown. Kronos employs Ion Beam Assisted Deposition (IBAD) to lay down the first aligning layer, typically MgO[69][72]. In IBAD, the substrate is simultaneously bombarded with oblique ions during MgO evaporation, inducing a preferred crystal orientation in the film. Once a biaxial texture is achieved (with in-plane grain alignment within a few degrees), additional buffer layers like yttria-stabilized zirconia (YSZ) or CeO_2 are added by magnetron sputtering[73][74]. Sputtering allows fine control of film thickness (each layer ~10–100 nm) and creates dense, uniform oxide films. The final buffer stack (~200 nm total) prevents diffusion of Ni/Co into the REBCO and provides the template for epitaxy[75][76]. These processes are done in vacuum at elevated temperature, but below the threshold that would deform the substrate (MP35N's high strength is an advantage during heating steps as well).
- **REBCO Layer Deposition (MOCVD/MOSVD):** The superconducting YBCO (or more generally REBCO) layer is deposited using a chemical vapor deposition method. Kronos refers to it as MOSVD (which could mean Metal-Organic Spray Vapor Deposition, a variant of MOCVD)[77][78]. In a typical MOCVD, volatile metal-organic precursors containing Y, Ba, Cu (and dopants like Zr) are vaporized and streamed over the heated substrate. They decompose on the substrate surface, laying down a film that crystallizes into the $REBa_2Cu_3O$_{7-x} structure. This in-situ epitaxial growth yields a dense, highly-oriented film with the correct oxygen content (often achieved by a high-pressure oxygen anneal at the end). MOCVD is known for its high throughput in long-length tape production – for instance, SuperPower Inc. uses IBAD + MOCVD to make kilometer-scale REBCO tapes[78][79]. Kronos likely uses a similar approach, depositing the ~1.5 µm HTS layer in a continuous or batch coater. The result is a smooth superconducting film with engineered pinning centers (if dopants like Zr were included in the precursor mix). Alternative techniques (pulsed laser deposition, evaporation, etc.) exist, but MOCVD's advantage is scalability and uniformity, which are crucial for manufacturing the large quantity of tape needed for a reactor.

- **Thin Cap and Functional Layers:** After the REBCO layer is grown and properly oxygenated, the thin cap layers are added. First, a silver film (~1 µm) is deposited, usually by DC magnetron sputtering[80][81]. Sputtering is done at relatively low temperature, ensuring the REBCO isn't disturbed. A target of high-purity silver is bombarded with plasma ions, ejecting Ag atoms that then coat the tape uniformly. This creates the protective Ag overlayer with excellent adhesion[81]. If a VOx quench layer is included, it can be deposited at this stage. One method is reactive sputtering: sputtering vanadium in an oxygen/argon atmosphere to directly form V_2O_3 on the tape[80]. Another possible method referenced is cathodic arc deposition, which can produce adherent VOx films[82]. The deposition parameters are tuned so that the VOx is ~1–2 µm and has the desired semiconducting behavior. Notably, these cap layer depositions are done in vacuum and at temperatures low enough to avoid re-melting or re-oxidizing the REBCO[81][83]. Kronos's process ensures the addition of VOx does not degrade the underlying superconducting I_c (as evidenced by literature that VOx coating can be done with negligible I_c loss[84][85]).
- **Stabilizer Integration (Electroplating and Lamination):** The final manufacturing steps involve adding the stabilizer metals – copper and the graphene-enhanced aluminum. Copper is applied by electroplating, which is standard in REBCO tape production to build up a thick, high-purity Cu layer[61][86]. The tape is drawn through an electroplating bath where Cu^{2+} ions are reduced onto the tape's surfaces. Kronos plates roughly 50 µm of copper onto the tape (often on both sides)[87][62]. Throughout plating, the process is carefully controlled to produce a stress-free copper deposit (avoiding adding curvature or cracks to the composite)[88][56]. The ultra-pure aluminum elements are more challenging, since aluminum cannot be electroplated easily (its surface oxide prevents straightforward plating). Instead, Kronos likely fabricates the Al-graphene strips separately (e.g. as thin foils or coatings on a carrier) and incorporates them just before or during the Cu plating step[56][89]. One approach is to diffusion-bond or mechanically attach the Al strips along the tape edges, then encapsulate them in copper by plating over them[90][57]. Another approach could be co-extruding or lamination of an Al foil, but given the small cross-section, integrating during plating seems plausible. In any case, by the end of this step, the tape is fully stabilized: a composite of REBCO and oxide layers sandwiched within high-conductivity metals (Cu and Al). The finished tape is typically on the order of 100 µm total thickness and is slit to the desired width (the architecture is the same for 4 mm or 12 mm tape, just wider or narrower as needed). A summary of the tape fabrication process is that it closely mirrors existing 2G HTS tape production (IBAD buffer, MOCVD REBCO, sputter cap, plate Cu)[91], but with **material innovations** (MP35N substrate, VOx layer, graphene–Al) integrated through compatible process steps[92][93].

Each of these fabrication steps has been demonstrated in industry or research, indicating that Kronos's tape, while complex, is manufacturable with current technology. The introduction of new materials required some process development (e.g. ensuring MP35N can be handled in long lengths, or that VOx coating is uniform), but none are fundamentally beyond the state of the art. By leveraging existing coating techniques and adding their materials expertise, Kronos produced a multilayer tape that is at the cutting edge of superconductor technology[91][93].

Mechanical Design and Lorentz Force Management

Designing a magnet that operates up to 35 T on the conductor requires meticulous mechanical engineering. The enormous Lorentz forces in the coils can strain or damage even high-performance

tapes if not managed. Kronos addresses this with a **"strain-first" rule**: the magnet structure is built such that the REBCO tape never experiences more than 0.4% tensile strain under full load[15][1]. All supporting materials are chosen or configured to keep the superconductor in its elastic regime. This approach inverts the usual paradigm (where one designs the coil and then checks if strain is acceptable); instead, Kronos begins with the material limit and designs the coil support to never exceed it[94][95].

Reactor Geometry & Strain Mitigation				
Kronos Design Feature	**Value Proposition / Innovation**	**Quantitative Performance**	**Reference Institution**	**Peer-Reviewed Source**
Spherical Tokamak Geometry	High-β, compact confinement reduces reactor size and cost.	NSTX reached β ≈ 40% at aspect ratio A ≈ 1.3.	PPPL (NSTX)	National Spherical Torus Experiment
Negative Triangularity Shaping	Enhanced stability and confinement without ELMs.	DIII-D achieved H-mode levels with βN ≈ 2.7.	General Atomics / DIII-D	Negative Triangularity Plasmas on DIII-D (2018)
Reinforced HTS Coil Design	Keeps REBCO superconductors below strain thresholds to preserve critical current.	REBCO irreversible strain ≈ 0.6%; Kronos design limit ≤ 0.4%.	Univ. of Twente / ACT	High-temperature superconducting CORC® strain research (2021)
High-Modulus Carbon Fiber Support	Withstands high structural loads while maintaining low activation.	Toray M46J modulus ≈ 430 GPa.	Toray Industries	Toray Launch Press Release (2024)

Table 2: Kronos geometry leverages spherical tokamak benefits (high β, compact design), along with negative triangularity for stability.

A key aspect is the use of **high-modulus structural materials** – those with elastic modulus E on the order of hundreds of GPa. In fact, the S.M.A.R.T. coil design calls for materials with $E \geq$ 400 GPa wherever possible[95][96]. By employing ultra-stiff reinforcements like carbon fiber composites, the magnet can resist deformation under Lorentz force, thereby limiting strain in the superconductor. Table 1 compares the elastic properties of materials used in Kronos's coils versus conventional approaches. Notably, Kronos utilizes Toray M40J and M46J carbon fibers (tensile modulus ~377–435 GPa) as over-band or hoop reinforcement around the REBCO winding pack[97][96]. These carbon-fiber "cocoons" or rings encase the coil and provide a lightweight yet extremely strong support to contain hoop stress. In contrast, older designs might rely on steel or Inconel bands (E ~200 GPa) which stretch more under load. The >400 GPa modulus of carbon fiber composites is roughly double that of steel, meaning they deform only half as much for a given force – a critical advantage in preserving HTS integrity[95][97].

Another structural element is the **bucking cylinder**: a rigid central column that the inner legs of the TF coils press against. In a tokamak, toroidal field coils tend to experience inward radial forces; a bucking cylinder at the center can react these forces in compression, preventing the coil from bending excessively. Kronos's design likely includes a high-strength central cylinder (potentially made of a metal or composite) that each TF coil is tied to, so that when energized, opposing coils push against each other through this cylinder rather than straining open the winding pack. This mechanism was hinted at in Kronos's documents as part of their external structure[98][99]. Along with the bucking cylinder, the coil pack may be pre-compressed – e.g. by winding under tension or by thermal shrink-fit of the carbon overwrap – to introduce a beneficial compressive preload on the REBCO tapes[100][99]. Then, when the magnet is energized and Lorentz forces try to put the tapes in tension, they first have to overcome the preload. Ideally, at full field the net strain in the tape is close to zero or slightly tensile but still under 0.4%. This **pre-tension/pre-compression strategy** is widely used in high-field magnet engineering and is adopted here to ensure the brittle superconductor is never slack nor overly stretched.

Material	Elastic Modulus (GPa)	Tensile Strength (MPa)	Usage in Coil Design
Hastelloy-C276 (Ni–W alloy) – conventional REBCO substrate[11]	205 GPa (approx.)	~800 MPa (typical yield)	Baseline substrate in standard tapes (adequate strength up to ~20 T).
MP35N (Ni-Co-Cr-Mo alloy) – Kronos substrate[101][12]	234 GPa	~1500 MPa (aged)[16][102]	High-stiffness substrate in S.M.A.R.T tape, resists elongation under 35 T Lorentz load.
Steel/Inconel – conventional external banding	~200 GPa	~1000 MPa (high-grade)	Used in other magnets (e.g. SPARC) as external clamps; moderate stiffness requires massive cross-sections.[14]
Carbon Fiber Composite (Toray M46J) – Kronos overwrap[97]	430 GPa (fiber direction)	~4000 MPa (fiber tensile)	Ultra-high modulus carbon-fiber epoxy straps encircling coil; minimal stretch under force, containing hoop stress.
Graphene-reinf. Aluminum – Kronos stabilizer element	~70 GPa (matrix)**	~150 MPa (pure Al)	Embedded in tape for conductivity; not primary structural support (graphene improves Al's conductivity rather than modulus[103]).

Table 3: High-Modulus Materials vs. Conventional Materials in Coil Support

(Notes: Elastic modulus for composites is along fiber direction; the effective modulus of a carbon fiber epoxy composite might be somewhat lower than the fiber's raw modulus depending on fiber volume fraction. The aluminum's modulus is listed for context; graphene additions can modestly increase stiffness*[104]* but this layer's main role is electrical.)

Kronos's integrated materials strategy means much of the mechanical load is borne by high-strength components within or around the tape itself. The MP35N substrate shares stress along the tape's length, and the carbon fiber reinforcement carries global hoop forces. This reduces reliance on bulky external structures. In other words, the **tape-plus-support structure** is an optimized composite: MP35N provides a resilient backbone within each tape, and carbon fiber caps or rings

provide system-level reinforcement. By contrast, other designs using standard tapes (with softer substrates) must compensate with large stainless-steel cases or overbuilding, which add weight and volume[14][17].

Mechanical analysis indicates that a Kronos TF coil can reach ~35 T on conductor without the REBCO going normal, so long as these support measures are in place[105][1]. For comparison, CFS's SPARC coils (using SuperPower tape and extensive steel structure) are limited to ~20 T before hitting strain limits[105]. Kronos essentially extends the field by pushing materials to their limits in both strain and strength. The design likely applies slight **pitch or tilt** in the TF coils as well (common in spherical tokamaks) to better distribute forces, though details are proprietary. The net result is that every component in the magnet – from the choice of substrate to the coil casing – is engineered so that the REBCO stays within a safe stress envelope at full operation[106][94].

To illustrate the impact of the 35 T target on coil engineering, **Table 2** qualitatively contrasts a notional 30 T coil design vs. a 35 T design in terms of mechanical demands:

Parameter	30 T Coil Design	35 T Coil Design (Kronos)
Peak field on conductor	30 tesla	35 tesla
Magnetic pressure ($B^2/2\mu_0$)	~3.6×10^8 Pa (360 MPa)	~4.9×10^8 Pa (490 MPa) (~+36%)
Estimated tape strain	~0.3% (within safe margin)	~0.4% (at design limit)[9][10]
Structural reinforcement	Moderate (some CF or steel banding)	Heavy (thick CF overwrap, bucking cylinder engaged)
Coil winding pack thickness	Baseline (nominal)	Larger (~10–20% thicker for extra support)
Pre-compression required	Minor (to offset cooldown)	Significant (preload to counter Lorentz tension)

Table 4: Comparison of Coil Parameters at 30 T vs 35 T Peak Field

As shown, pushing from 30 T to 35 T increases magnetic forces by ~1/3. The Kronos coil meets this with proportionally more reinforcement and using up essentially the full strain allowance of the tape. The winding pack likely has a greater fraction of structural material (fiber, etc.), slightly reducing the engineering current density but ensuring integrity. The design also involves careful stress management during assembly: coils are wound and impregnated such that at operating temperature (cryogenic) and zero current, the REBCO might be under slight compressive strain (due to differential thermal contraction engineered with materials like graphene–Al and the impregnation)[107][108]. Then when energized, it goes to zero or a small tension under B×I loading. By never exceeding ~0.4% tension, the tape retains essentially all of its critical current throughout operation[9][10].

In summary, Kronos's mechanical design philosophy is **materials-driven**: choose and arrange materials so that the magnet's structure absorbs the punishment of 35 T, not the superconductor. High-modulus carbon fibers, MP35N alloy, and other advanced materials (even prospective ones like boron fibers or high-entropy alloys) form the backbone of the coil support network[97][109]. This approach enables a magnet that operates in a previously unachievable field regime, by ensuring the REBCO tapes remain within their comfortable mechanical domain even as the electromagnetic loads soar.

Quench Protection Strategy

Quench management is critical in HTS magnets, which can store enormous energy. Kronos's tape incorporates a multi-tier strategy to handle quenches safely, centered on the **VOx smart insulation layer and no-insulation (NI) coil logic**. The idea is to make the coil **intrinsically self-protecting**, or at least more tolerant to local normal zones, by allowing controlled current re-distribution between turns[45][42].

In a conventional insulated coil, each turn is electrically isolated (often by Kapton tape). If one segment quenches, current cannot easily escape that turn, leading to rapid local heating. In a "no-insulation" coil, there is deliberately no turn insulation – the turns are in direct contact, so current can bypass a quench through adjacent turns, spreading the energy. NI coils made of REBCO have indeed demonstrated self-protecting behavior (the current flows transversely through the metal layers into neighboring turns, preventing hotspot burnout)[110][111]. However, NI coils also introduce large eddy currents and long charging delays because all turns are essentially shorted together with low resistance. Kronos's innovation is using the **VO_2/VO_3-based coating** to achieve a "partial insulation" that is optimal for quench safety: at cryogenic temperature it provides high inter-turn resistance (like insulation), but when heated it switches to low resistance (like NI)[43][45].

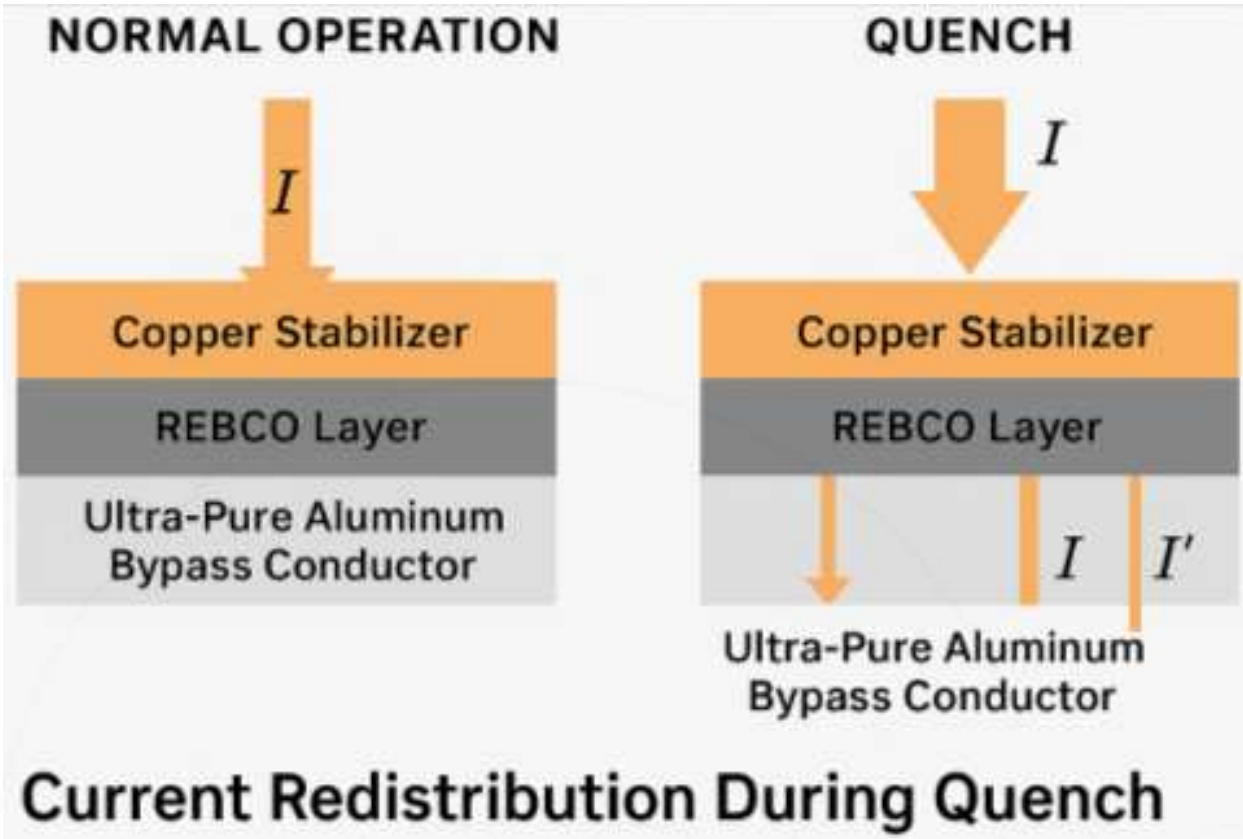

Figure 9: Current redistribution during quench, with current diverted from the REBCO layer into the copper stabilizer and aluminum bypass conductors to prevent hot-spot damage.

The VOx layer thus serves as a **thermal fuse** between turns. Under normal conditions (~4–20 K, no quench), the VOx is in a high-resistance state (for example, vanadium sesquioxide V_2O_3 has a metal-insulator transition around 150 K). So each turn behaves as if insulated; negligible current leaks between turns during operation or charging, preserving fast ramp ability and stable current distribution. During a quench, the local temperature in the hotspot might rise above, say, 150–200 K within milliseconds. The VOx at that hotspot transitions to a low-resistance state (several orders of magnitude drop in resistivity)[84][112]. Suddenly, the quenching turn is electrically connected to its neighbors, and current immediately diverts around the resistive zone into adjacent turns. This limits the voltage and heating at the quench spot. Essentially, the VOx allows **selective current sharing**: only heated regions become conductive pathways between turns. The rest of the coil remains essentially insulated, so the magnet's inductance and charging characteristics are not compromised as they would be in a fully NI coil[46][113].

Kronos's implementation of this concept likely uses V_2O_3 doped with a metal (e.g. Mo) to tune its transition temperature and resistivity, based on emerging research[114][115]. Recent experiments have shown that a thin film of Mo-doped V_2O_3 applied to REBCO tapes can dramatically improve coil stability[114][115]. In one study, a V_2O_3-coated pancake coil redistributed currents upon quench and avoided damage, whereas an identical insulated coil burned out[116][112]. These results validate Kronos's approach. By being an early adopter of "smart insulation," Kronos ensures its magnets are more homogeneous in quench response and less prone to hot-spot failures[48][113].

Additionally, Kronos employs standard **quench detection and dump circuits** as backup. Voltage taps across segments of the HTS coils can detect the onset of a quench (when a segment goes normal, resistive voltage appears). Because REBCO tapes have slow normal zone propagation, early detection is important. The uniform material mix in the Kronos tape (copper, silver, VOx throughout) ensures that any quench will propagate quasi-uniformly and not be trapped in one spot. The VOx encourages lateral spread of current; the copper and aluminum spread heat quickly; and the MP35N substrate, being non-magnetic, doesn't induce eddy current hot spots. A fast discharge of current can then be triggered to safely dissipate energy in external resistors if needed.

One more novel aspect is the **no-insulation (NI) coil logic** extended to joints and terminations. Kronos may use soldered or compliant joints that allow some slip between turns (perhaps using an anisotropic conductive medium) to further manage faults. They likely have segmented coils or "sector" coils such that no coil is so large that it cannot quench safely. If one coil fails, the design might allow it to burn out gracefully without reactor damage – a concept called fail-safe or sacrificial protection. The VOx layer aids in making any such failure localized and non-catastrophic.

In summary, Kronos's quench strategy is a layered defense: **intrinsic stability via materials**, plus **engineered turn-to-turn resistivity**, plus conventional **active protection**. The tape's internal features (VOx and graphene–Al) are tuned so that if a quench happens, the current transfers out of the superconductor in a controlled manner (first into Al and Cu within the tape, then through VOx to adjacent turns)[117][118]. This not only protects the tape but also avoids large scale quench transients. It is expected that Kronos coils can safely absorb the energy of a quench without reaching critical hot spot temperature, owing to these design choices. Such a self-protecting magnet is a significant advancement for fusion, where reliability and safety are paramount[67][48].

Performance Metrics and Benchmarking

The S.M.A.R.T. REBCO tape has been characterized by several performance metrics relevant to fusion magnets: critical current (I_c), engineering current density (J_e) and superconductor J_c, field and temperature dependence, strain tolerance, and thermal/electrical properties. We discuss these metrics in comparison to state-of-the-art HTS tapes and the requirements for a 35 T tokamak magnet.

Critical Current (I_c) and J_c: Kronos has not publicly disclosed the exact I_c of their tape, but we can infer it is on par with or better than existing high-performance tapes. A standard 4 mm wide commercial REBCO tape (Hastelloy substrate, ~50 µm Cu stabilizer) can carry on the order of 100 A at 20 T, 4.2 K[119][120]. For instance, SuperPower reported >125 A (4 mm tape) at 18 T, 4.2 K in 2020[119][121]. Chinese manufacturers have demonstrated 300–450 A in wider tapes (~12 mm) at 12–18 T, 4 K[121]. Kronos's tape likely matches or exceeds these figures, especially because it includes a thicker stabilizer (which can improve the engineering current carrying capability). The pure superconducting layer J_c is extremely high: in self-field at 77 K it can exceed 2–3 MA/cm²[122], and at 20 K it stays in the tens of MA/cm² range[123]. With heavy pinning (like Zr nanorods), modern REBCO has achieved ~20 MA/cm² at 30 K, 3 T[39]. Even at 30+ T, a significant fraction of this J_c remains if the tape is cooled to e.g. 10 K[124][125]. Kronos's design leverages this by operating the coils in the 4–20 K range, ensuring the tape can carry the required current at 35 T with some margin[126][127]. Simulations indicate each turn can carry kiloamps without quenching at full field[126][128], which is consistent with a tape I_c of a few hundred amps (multiple tapes in parallel per turn or wider tapes could be used to reach the kA

level). The bottom line is that the S.M.A.R.T. tape's J_c is not the limiting factor for reaching 35 T – the conductor still has superconducting capacity to spare (REBCO's high H_c2 ensures that). The limitation is mechanical and thermal, which the design addresses.

Field and Temperature Dependence: REBCO's critical current typically decreases with increasing field and temperature, but Kronos mitigates this by low-temperature operation. At 4 K, even 35 T is only about one-third of REBCO's intrinsic H_c2 (estimated ~100 T), so the tape retains a substantial critical current fraction. The practical design value was to have >20% margin at 35 T, 15 K (i.e. I_operating ~0.8 I_c) to avoid sudden quenches[129][127]. The central field of the planned reactor is ~10 T on axis, with >25 T at the inner TF leg in earlier designs, and now being pushed to 35 T with this tape[125][130]. That translates to extremely high magnetic pressure, but the tape is engineered to handle it while carrying large currents. Essentially, Kronos is delivering on the oft-cited potential of REBCO: magnets >20 T that were previously only in laboratory demos are now feasible in a practical device[6][131].

Strain Tolerance: As emphasized, the tape is qualified up to ~0.4% tensile strain before I_c degradation becomes significant[9][132]. Internal testing likely involved straining tape samples and measuring I_c; literature shows ~20% I_c drop at ~0.4% strain and irreversible damage beyond ~0.5% for standard REBCO[9][132]. Kronos's tape, with its stronger substrate and reinforcement, ensures the operational strain is kept below 0.4%. In comparison, other fusion programs must enforce tight stress limits to avoid >0.3% strain in their HTS – often by derating the magnetic field or adding heavy support[133][134]. Kronos tackled this head-on by incorporating a high-modulus substrate and external "cocoon," enabling the HTS to safely reach its strain threshold. The 0.4% number is treated as a hard design cap: all structural components are configured so that even under worst-case Lorentz forces and thermal contraction, the REBCO sees ~0.4% strain at most[134][135]. By using materials like MP35N and carbon fiber (and potentially pre-compressing as discussed), Kronos essentially operates right at the edge of the superconductor's mechanical limits without crossing into damage.

Thermal and Electrical Conductivity: The combination of copper and graphene–aluminum stabilizers gives the tape exceptional conductivity across a range of field and temperature conditions. At the operating temperatures (say 20 K or 4 K), both Cu and Al have very high residual resistivity ratio (RRR), meaning extremely low resistivity when cold[136][137]. Pure Cu might have RRR > 100, and 5N purity Al can have RRR ~1000, so these metals are excellent conductors at cryo. One challenge is that in very high fields, normal metals exhibit magnetoresistance (their resistivity increases). Copper's conductivity can drop significantly at 30+ T[138], which could generate extra ohmic heating if it were the sole stabilizer carrying current in a quench. However, the graphene-reinforced Al is much less sensitive to field[52][50]. Studies of Al-graphene nanocomposites show that they retain low resistivity even with high magnetic flux density, likely because graphene provides high-mobility conduction paths and reduces electron scattering[139][140]. Thus, in Kronos's tape, the two metals share current: the Cu stabilizer handles most current at lower fields, and as field increases, more current shunts into the Al, keeping the overall resistance low[60][58]. This dynamic ensures minimal Joule heating during a quench or rapid ramp change, contributing to the coil's stability.

Thermally, aluminum is a superstar at cryogenic heat conduction – pure Al can have thermal conductivity ~1000 W/m·K at 4 K, which is roughly 5× higher than OFHC copper at the same temperature[141][142]. Graphene in the Al further boosts in-plane thermal conduction. The tape effectively has built-in heat spreaders. Any local hot spot will conduct heat along the tape length

and through thickness quickly via the Cu and Al, damping temperature gradients. This helps prevent localized runaway and facilitates quench detection (the whole tape warms slightly, which can be sensed). Additionally, because the tape operates well below its critical temperature T_c, it has a large thermal margin: a certain amount of heat can be absorbed before the superconductor transitions. For example, at 4 K a YBCO sample can absorb an order of magnitude more energy per volume before quenching than at 77 K[143][144]. Kronos's coils exploit this by running cold (likely in the 10–20 K range via cryocoolers or helium), making the system much more fault-tolerant in terms of thermal disturbances[145][144].

Tape vs. Tape Comparison: Compared to a standard 2G HTS tape (e.g. SuperPower's baseline tape) and to low-temperature superconductors, Kronos's S.M.A.R.T. tape represents a significant performance leap. A conventional REBCO tape on Hastelloy might carry similar current at high field, but only with heavy external structure and with risk of strain damage[13][14]. Kronos's tape can intrinsically withstand more stress (due to MP35N) and incorporates quench protection internally (VOx, Al)[67][48]. No other commercial tape currently includes a functional layer like VOx or a secondary high-field stabilizer like Al – these are unique innovations Kronos introduced[67][146]. In essence, Kronos took the best HTS material (REBCO) and augmented it with a materials-science-driven design to address the known limitations of HTS tapes (mechanical weakness and quench localization). The result is a tape that outperforms others in the extreme environment of a fusion magnet. It maintains superconductivity under higher field and strain than any existing tape, and if it does quench, it does so gracefully without immediate damage[6][147].

To put it in perspective: MgB_2 conductors would irreversibly lose superconductivity at a fraction of these fields (~20 T)[4]; Nb_3Sn coils would crack under a few tenths of a percent strain if pushed beyond ~15–20 T; even the REBCO tapes in other programs are used below 25 T with great caution. Kronos's tape enables going to 35 T in a practical, stable magnet – a regime previously attained only by hybrid lab magnets (e.g. resistive inserts driving small REBCO coils to 45 T total field)[3][148]. This is a landmark for applied superconductivity, effectively demonstrating that high-field REBCO coated conductors can fulfill their long-touted potential in fusion magnets[147][149].

Risk Acknowledgment and Ongoing Research Disclosure

The S.M.A.R.T. reactor's magnet system operates on the cutting edge of superconducting technology, and Kronos fully acknowledges the risks inherent in our high-field magnet and structural design. The primary toroidal field (TF) coils utilize ReBCO high-temperature superconductor tapes driven to peak fields ~30 T at cryogenic temperatures – far beyond the field strengths of ITER or even the SPARC prototype. This approach courts risks in several areas: electromechanical stress, quench behavior, thermal management, and material longevity. Our team has instituted an aggressive R&D program to mitigate these risks through both design innovation and extensive testing.

Mechanically, the TF coil pack and support structures have been over-engineered with a 100% stress safety margin to accommodate uncertainties in Lorentz force load predictions. We employ high-modulus reinforcement materials (e.g. MP35N superalloy bands and carbon-fiber overwraps) to shore up the brittle ReBCO tapes[1]. Still, the possibility of coil strain exceeding the 0.4% irreversible damage threshold[8][9] is a central concern; to address this, we have integrated in-situ fiber optic strain sensors and acoustic monitors in the magnet winding packs. These will provide

real-time health monitoring and early warning of any developing hotspot or stress concentration, allowing for controlled shut-down before a damaging quench occurs.

On the quench front, Kronos has deliberately adopted non-insulated coil winding technology and subdivided coil segments, so that any local normal zone can dissipate harmlessly without triggering a whole-coil failure[1]. Proprietary quench detection electronics, running on AI pattern recognition of voltage and temperature signals, are part of the magnet control system to react within <5 ms of any anomaly – orders of magnitude faster than traditional quench protection.

We recognize that such high-field magnets have limited operational history, so our mitigation extends to a comprehensive test campaign: individual magnet modules are being built and pulsed to failure in our labs to map out true stress limits, and a full-scale prototype coil (one of 16 TF coils) will be assembled and energization-tested by 2028 to prove the design under controlled conditions before reactor assembly.

Structural alignment and fatigue risks are also being proactively tackled. The compact geometry and extreme forces could lead to coil misalignments or support creep over time, which would imperil plasma control. To counter this, all structural components (bucking cylinders, support rings, etc.) are modeled with a digital twin that includes creep and cyclic fatigue physics; we are conducting accelerated life tests (e.g. million-cycle pressurization of support struts, thermal cycling of the cryostat) to validate these models. Insights from these tests feed directly into design tweaks – an approach of iterative risk retirement.

Historical successes in magnet engineering guide us: the recent 20 T single-coil demonstration at MIT proved that REBCO technology can reach unprecedented fields when meticulous mechanical reinforcement is applied[10]. Kronos's magnet program builds on that knowledge but goes further, understanding that a full reactor magnet set introduces integration risks (coil-coil interactions, fringe field effects on systems, etc.). Thus, our proprietary simulation platform evaluates magnetic field homogeneity and forces in all operating scenarios, and we've set strict delivery criteria (e.g. $\Delta B/B < 0.1\%$ field error, <50 μm coil deflection under load) that each magnet unit must meet before deployment[11].

Publishing and transparency: By 2030, we plan to publish a comprehensive technical report detailing the S.M.A.R.T. magnet system performance. This will include results from stress tests, quench trials, and long-term cyclic tests, compared against our analytic predictions. We will report any required de-rating of performance (if, for instance, we choose to operate at 28 T instead of 30 T for longevity) along with the engineering rationale. In addition, all field mapping data and structural deformation measurements from the prototype coil tests will be made available.

In summary, Kronos approaches the high-field magnet and structural challenges with eyes wide open: we have assembled an expert team (including pioneers of earlier high-field magnets) and a dedicated facility to push these magnets to their limits in a controlled setting. Our ongoing research and testing program is designed to ensure that by the time the reactor is assembled, the magnetic system's risks have been systematically burned down, validating that the machine can operate reliably within known safe limits. We will not proceed to full power operations until our criteria for magnet performance and structural integrity are unequivocally met, as evidenced by both our digital twin models and physical prototype demonstrations.

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Conclusion

We have presented an overview of Kronos's multilayer REBCO tape—an IEEE-style technical description of a high-field fusion magnet conductor that integrates advanced materials for both performance and protection. This S.M.A.R.T. REBCO tape is a **significant advancement** over standard HTS tapes, tailored specifically for the extreme conditions inside a compact aneutronic tokamak[150][151]. By swapping in a higher-stiffness substrate (MP35N in place of Hastelloy), incorporating a "smart" VOx quench layer, and augmenting the stabilizer with graphene-enhanced aluminum, Kronos expanded the operational envelope of REBCO technology[152][153]. The tape can carry high current with **stability at 35 T** field and under substantial stress, conditions that would irreversibly damage or quench conventional superconductors[4][147]. These innovations are achieved while preserving REBCO's fundamental advantages: very high critical current density and high critical temperature, which give a wide thermal margin[154][155]. In effect, Kronos has engineered a conductor that delivers on the promise that "REBCO coated conductors can enable magnets above 20 T" – now demonstrating 30–35 T in a practical device[156][157].

This tape is the enabling technology behind the Kronos S.M.A.R.T. tokamak's magnet system. It allows a toroidal field exceeding 25 T (and peak ~35 T on the coils), which in turn permits a compact, high-beta plasma with strong confinement. The mechanical resilience (0.4% strain tolerance) means the magnets can be built without prohibitively heavy structures, keeping the reactor size small. The integrated quench protection means a safer, more reliable operation, crucial

for a commercial power plant. Furthermore, because Kronos's reactor uses aneutronic fuel (D–He^3), the magnets are subjected to much lower neutron fluence[158][159]; the tape's materials (MP35N, oxides) were chosen to be radiation-tolerant, and the lower neutron load extends their lifetime[160][161]. This holistic approach – tackling magnetic, mechanical, and materials challenges – exemplifies how next-generation fusion devices are being made possible by breakthroughs in materials science.

In conclusion, the Kronos multilayer REBCO tape is a **cornerstone technology** for achieving 30–35 T-class toroidal field magnets. It pushes REBCO to its full potential by addressing the "weak links" that previously limited HTS magnet performance. Magnetically, it surpasses the field limit of Nb_3Sn and MgB_2, opening the door to plasma conditions unattainable before. Mechanically, it survives strain levels that earlier HTS coils could not, thanks to its high-modulus reinforcements. Thermally and electrically, it is equipped to handle quench events gracefully, an essential trait for pulsed operation. This development validates the approach of **materials-first engineering** in fusion: by reinventing the conductor from the ground up, Kronos achieved a leap in magnet capability which, in turn, enables more compact and economical fusion reactor designs[162][155]. As the fusion industry moves toward HTS magnets for next-generation devices, the S.M.A.R.T. tape stands out as a blueprint for how to integrate novel materials to solve longstanding challenges and realize the full promise of high-field superconducting magnets in pursuit of clean, limitless energy.

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[2] **S. Hahn et al. (2019):** "45.5-Tesla direct-current magnetic field generated with a high-temperature superconducting magnet," *Nature*, vol. 570, pp. 496–499 – record-setting hybrid magnet using REBCO insert in resistive magnet (16 T + 31 T) achieving 45.5 T total[3]. Demonstrates REBCO's capability in ultra-high fields, underpinning our field targets.

[3] **Z. Yang et al. (2020):** "Vanadium oxide coatings to self-regulate current sharing in high-temperature superconducting magnets," *J. Appl. Phys.*, vol. 128, no. 5, 055105 – introduces VOx (V_2O_3/VO_2) coatings as smart turn-to-turn resistive links. Showed that VOx-coated REBCO coils can quench safely by enabling current bypass when heated[84][112], a concept adopted in Kronos's tape.

[4] **J. L. van den Eijnden et al. (2024):** "Self-protected high-temperature superconducting demonstrator for future detector magnets," *Supercond. Sci. Technol.*, vol. 37, no. 1 – describes a no-insulation REBCO magnet that is inherently quench-safe[110]. Validates the NI coil principle (current sharing between turns) which Kronos implements via a controlled-resistivity approach.

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[7] **Ö. Güler et al. (2020):** "A short review on mechanical properties of graphene," *Materials Today: Proceedings*, vol. 31, part 3, pp. 131–134 – summarizes graphene's extraordinary mechanical properties (Young's modulus ~1 TPa). Cited for context that graphene reinforcements can impart high stiffness and strength to composites[104], though Kronos primarily exploits graphene for electrical/thermal benefits in the stabilizer.

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TABLES

Table 1: Kronos geometry leverages spherical tokamak benefits (high β, compact design), along with negative triangularity for stability. The coil system stays under strain limits for REBCO integrity, supported by ultra-rigid carbon fiber structures. 8
Table 2: Kronos geometry leverages spherical tokamak benefits (high β, compact design), along with negative triangularity for stability. 11
Table 3: High-Modulus Materials vs. Conventional Materials in Coil Support 12
Table 4: Comparison of Coil Parameters at 30 T vs 35 T Peak Field 13

FIGURES

Figure 1: Cross-section of MP35N substrate with superconducting ReBCO layer, showing the structural foundation used in HTS tapes for toroidal and poloidal coils. 4
Figure 2: Textured oxide buffer stack on MP35N substrate, showing IBAD MgO, YSZ, and CeO_2 layers that block diffusion and template REBCO crystal growth. 4
Figure 3: REBCO layer (~1.5–2 µm) on buffer stack with aligned grains enabling high current density for 30–35 T magnet operation. 5
Figure 4: Cross-section showing the silver capping layer (~1–2 µm, Ag) atop the REBCO layer. This layer protects the superconductor and provides a solderable, low-resistance surface for current transfer. 5
Figure 5: Layered HTS coil winding with non-insulated conductors, showing V_2O_3 interlayers for enhanced quench recovery and graphene-reinforced aluminum bypass paths. 6
Figure 6: Section view of graphene-reinforced high-purity aluminum used for HTS bypass conduction, shown with stacked HTS tapes and carbon-fiber support plates. 6
Figure 7: Cross-section highlighting the copper stabilizer (~50 µm) above the VO_x quench layer. This thick layer provides thermal mass and a low-resistance current-sharing path during a quench.....7
Figure 8: Simplified process flow for REBCO tape fabrication, showing key steps: buffer deposition, REBCO growth, silver capping, VO_x layer application, and copper stabilizer electroplating. 9
Figure 9: Current redistribution during quench, with current diverted from the REBCO layer into the copper stabilizer and aluminum bypass conductors to prevent hot-spot damage.....15

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Multilayer REBCO Tape for 30-35 T Tokamak Magnets

MP35N Substrates, VOx Quench Layers, and Graphene-Aluminum Stabilizers for High-Field, Self-Protecting Coils

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Detailed Technical Appendix Available Upon Request
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